

Ritesh Vishwakarma

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EDUCATION

GAUTAM BUDDHA UNIVERSITY

COMPUTER SCIENCE ENGINEERING

INT. DUAL DEGREE

B.TECH + M.TECH | 2022

Gr. Noida, UP, India

CGPA: 8.0/10

SKILLS

PROGRAMMING

• Python • SQL

PYTHON STACK

• FastAPI • Pandas • NumPy
• Pytest • Twilio • Pipecat

GENERATIVE AI & NLP

• LangChain • LangGraph • CrewAI
• OpenAI • Gemini • HuggingFace
• LangSmith (Familiar) • Arize Phoenix
• RAG • Prompt Engineering
• MCP • Multi-Agent System

DATABASES

• MySQL • MongoDB • Redis
• Storage: **Blob** storage, **S3** bucket
• Vector DBs: Chroma, Weaviate, Faiss

DEVOPS & TOOLS

• Git • Docker • CI/CD
• Nginx • RabbitMQ
• Azure services • Azure AI Foundry
• Kubernetes (Familiar)
• Terraform (Basic)
• Claude Code • GitHub Copilot

CERTIFICATIONS

• Agent Systems with CrewAI
• LangChain & LangGraph
• Generative AI

ACHIEVEMENTS

• Designed voice agent featured at tech showcase
• Recognized for 40% latency optimization in APIs

EXPERIENCE

SENIOR SOFTWARE ENGINEER | MAR 2022 - PRESENT COMPANY: NAGARRO SOFTWARE PVT LTD

- Architected **scalable microservices** (FastAPI, Docker) utilizing **RabbitMQ** (topic producer/consumer) for queue-based asynchronous task distribution, serving **1000+ daily queries** with **99.9% uptime**.
- Architected an **LLM Gateway proxy** supporting semantic caching, rate-limiting, and multi-provider failover (Gemini to Claude) to ensure high availability.
- Led development of **RESTful APIs** with **JWT/OAuth2** (FastAPI, Redis), achieving 350-500ms response times and **reducing API latency by 40%**.
- Built **RAG-based search systems** using CrewAI, Weaviate, and LLMs, achieving **92% accuracy** across 1000+ documents (semantic search improved context by 35%).
- Developed a real-time **voice agent** using Pipecat and Twilio, enabling natural phone conversations with speech-to-text/text-to-speech pipelines.
- Developed an **NL2SQL engine** (CrewAI, SQL Server/MySQL) enabling non-technical users to query databases, handling **1000+ daily queries**.
- Engineered a **SharePoint integration pipeline** (Microsoft Graph API, Python), processing 10-15 documents daily into **Weaviate vector databases**.
- Integrated **Arize Phoenix** for production LLM observability, tracing token usage, latency bottlenecks, and agent tool execution.
- Created **CI/CD pipelines** (Azure Pipelines, Docker) reducing deployment times by **60%** with **85%+ code coverage** via pytest.

KEY PROJECTS

VOICE HIRING AGENT

- Built a voice-based AI hiring agent (FastAPI, LangChain) that automates candidate shortlisting via an LLM matching pipeline.
- Integrated Twilio and Pipecat to automatically call candidates, conduct conversational screening, and schedule interviews.
- Saved hours of manual recruiting effort by automating repetitive screening and scheduling tasks.

UNIFIED MARKETPLACE AGENT

- Developed a multi-agent system (LangGraph, Gemini) consolidating airport passenger services into a unified conversational assistant.
- Implemented dynamic multi-tenant configuration for catalog isolation, menu setups, and pricing across multiple airport entities.
- Integrated unified cart checkout and GA4-compliant ecommerce funnel event logging to track user drop-offs.

UNIFIED KNOWLEDGE & DATA RETRIEVAL PLATFORM

- Engineered a unified platform combining semantic document search (RAG) and database querying (NL2SQL) using CrewAI and Weaviate.
- Implemented hybrid keyword/semantic search, citation page tracking, and Redis caching for optimized sub-second responses.
- Added a real-time voice streaming interface via Pipecat and Twilio, enabling telephony-based query interactions.